Hedging the Treasury Lock

Working notes on Pucci (2019)

Reference paper: M. Pucci, Hedging the Treasury Lock, *Banca IMI SpA, Milan, August 19, 2019. SSRN working paper no. 3386521. Available at* https://ssrn.com/abstract=3386521.

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1 Trade and Notation

A *Treasury Lock* (T-Lock) is a pre-hedging derivative used by corporate issuers to lock the Treasury-rate component of an upcoming bond/loan issuance. At expiry t_e (typically $\leq$ 6M, occasionally 12M), the *long* party receives, on notional N,

$$\text{Payoff}_{t_e} \;=\; a\,N\,\text{RiskFactor}_{t_e}\big(\text{IRR}_{t_e} - L\big), \qquad a = +1 \text{ (long)},\; -1 \text{ (short)}, \tag{1}$$

where:

- L – the *Locked Yield* (strike), normally chosen close to the spot IRR_t at trade date;
- IRR_{t_e} – the internal rate of return at expiry of the prescribed Security (the *Standard T-Lock* fixes the security at trade date; the *Then-Current T-Lock* uses the on-the-run benchmark as observed at t_e);
- $\text{RiskFactor}_t \equiv -\partial P_t(y)/\partial y\big|_{y=\text{IRR}_t} = -10{,}000 \cdot \text{DV01}_t$, the bond's price sensitivity to a unit yield shift.

A long T-Lock is economically equivalent to synthetically shorting the Security between t and t_e.

2 Pricing Formulations

2.1 Bond price as function of yield

The T-Lock contract prescribes the simply-compounded bond pricing function. Denote $t_0 \leq t \leq t_1 < \cdots < t_n$ with $\alpha_1, \ldots, \alpha_n$ the accrual factors and c the coupon rate:

$$P_t(y) = \prod_{i=1}^{n} \frac{1}{1+\alpha_i y} \;+\; c \sum_{i=1}^{n} \alpha_i \prod_{j=1}^{i} \frac{1}{1+\alpha_j y}, \tag{2}$$

and within a coupon period ($t_0 < t \leq t_1$):

$$P_t(y) = \left(\frac{1}{1+\alpha_1 y}\right)^{\frac{t_1 - t}{t_1 - t_0}} \left[\prod_{i=2}^{n} \frac{1}{1+\alpha_i y} + c \sum_{i=2}^{n} \alpha_i \prod_{j=2}^{i} \frac{1}{1+\alpha_j y}\right]. \tag{3}$$

IRR_t is implicitly defined by $P_t(\text{IRR}_t) = P_t^{\text{mkt}}$.

For the analytical study of greeks Pucci uses the continuously-compounded analogue (Hull, Taleb)

$$P_t(y) = e^{-(t_n - t)y} + c \sum_{t_i > t} \alpha_i\, e^{-(t_i - t)y} \equiv P_t^{(0)}(y) + c\, A_t(y), \tag{4}$$

splitting redemption $P_t^{(0)}(y) = e^{-(t_n-t)y}$ from coupon annuity $A_t(y) = \sum_{t_i>t} \alpha_i e^{-(t_i-t)y}$. The k-th yield-derivatives are

$$D_y^k[P_t] = (t-t_n)^k e^{-(t_n-t)y} + c \sum_{t_i>t} \alpha_i (t-t_i)^k e^{-(t_i-t)y}, \tag{5}$$

yielding the alternating signs $D_y[P] < 0$, $D_y^2[P] > 0$, $D_y^3[P] < 0$ that drive everything below.

2.2 Arbitrage-free pricing

Under the t_e-forward measure (default risk on the Treasury ignored),

$$v_t \;=\; a\, D_{t,t_e}\, \mathbb{E}_t^{t_e}\big[\mathrm{RiskFactor}_{t_e}\big(\mathrm{IRR}_{t_e} - L\big)\big]\,. \tag{6}$$

Direct evaluation requires a stochastic model for the bond yield curve – substantially heavier machinery than warranted by the typical short tenor and quality of issuer. Section 3 justifies the desk's use of a forward-contract proxy.

2.3 Linearisation: T-Lock as price-gap

Linearising the bond function at IRR_{t_e}:

$$\mathrm{RiskFactor}_{t_e}\big(\mathrm{IRR}_{t_e} - L\big) = -D_y[P_{t_e}]\big|_{y=\mathrm{IRR}_{t_e}}\big(\mathrm{IRR}_{t_e} - L\big) \;\approx\; P_{t_e}(L) - P_{t_e}(\mathrm{IRR}_{t_e}) = P_{t_e}(L) - P_{t_e}^{\mathrm{mkt}}. \tag{7}$$

The T-Lock therefore measures, to first order, the gap between the bond's market price and its price computed by discounting flows at rate L.

3 The Forward-Contract Proxy

3.1 Booking convention

Approximation (7) suggests booking a long T-Lock as a *short* forward contract on the same security struck at

$$K \;=\; P_{t_e}(L) \;\approx\; \mathrm{ForwardPrice}_t(t_e) + \mathrm{RiskFactor}_{t_e}\big(\mathrm{IRR}_t - L\big). \tag{8}$$

This is the dominant desk practice when the T-Lock typology is not available in the front-office system.

3.2 Overhedge error: rigorous bound

Define the Taylor remainder

$$R_1(y) \;\equiv\; P(y) - P(L) - D_y[P](L)\,(y-L). \tag{9}$$

Proposition 1 (Global overhedge)**.** *For $a = +1$, the linear proxy strictly overhedges the T-Lock; equivalently, $R_1(y) \geq 0$ for all $y \geq 0$, with equality iff $y = L$.*

Proof. R_1 is twice differentiable with $R_1(L) = R_1'(L) = 0$ and $R_1''(y) = D_y^2[P](y) > 0$ from (5). Hence L is the unique minimum of R_1 on $\mathbb{R}_+$, value zero. So $R_1 \geq 0$, i.e. $P(y) \geq P(L) + D_y[P](y-L)$, and the proxy $P_{t_e}(L) - P_{t_e}(\mathrm{IRR}_{t_e})$ majorises the exact payoff. □

Proposition 2 (Quantitative bound)**.** *For all y in a neighbourhood I of L,*

$$|R_1(y)| \;\leq\; \tfrac{1}{2}\, M\, |y-L|^2, \qquad M = \max_{y\in I} \big|D_y^2[P](y)\big|. \tag{10}$$

For non-negative yields the worst-case bound is attained at $y = 0$:

$$M \;\leq\; (t_n - t)^2 + c \sum_{t_i>t} \alpha_i (t_i - t)^2. \tag{11}$$

Numerical sanity (10Y benchmark). Substituting $t_n - t = 10$, $c \sim 3\%$, $\alpha_i \sim 0.5$ gives $M \sim 150$. With a 6M expiry and a $\pm 30\%$ yield move (an upper bound), the dollar overhedge error is $\frac{1}{2} \cdot 150 \cdot (0.003)^2 = 6.75 \times 10^{-4} \approx 0.07\%$ of notional – i.e. a few basis points, dominated by the spread-based hedge cost.

4 Greeks of the T-Lock

Let $g(y) = -a\, D_y[P_t]\,(y - L)$ (the analytic payoff in y).

4.1 First and second derivatives in yield

$$D_y[g] = -a\big[D_y^2[P]\,(y - L) + D_y[P]\big], \tag{12}$$

$$D_y^2[g] = -a\big[D_y^3[P]\,(y - L) + 2\,D_y^2[P]\big]. \tag{13}$$

4.2 Delta (in price)

By the chain rule, $D_P[y] = 1/D_y[P] < 0$ (yield/price are order-reversing). Hence

$$D_P[g] \;=\; D_y[g] \cdot D_P[y] \;=\; \frac{D_y[g]}{D_y[P]}. \tag{14}$$

For $a = +1$ and $|y - L|$ small, (12) gives $D_y[g] > 0$, while $D_y[P] < 0$, so

$$\boxed{\Delta \;\equiv\; D_P[g] \;<\; 0 \;\text{ for a long T-Lock}} \tag{15}$$

i.e. the long T-Lock is short the bond's price, consistent with synthetic shorting via the forward contract.

Lemma 1 (Sharp delta-sign threshold)**.** *For $a = +1$, $\Delta < 0 \iff y < L - D_y[P]/D_y^2[P]$, equivalently $y - L < -D_y[P]/D_y^2[P] > 0$. The threshold exceeds L, so Δ remains negative for all y at or below L and well beyond it; sign flips occur only in extreme high-rate scenarios.*

4.3 Gamma (in price)

Differentiating (14) once more, using $(f^{-1})''(z) = -f''(f^{-1}(z))/[f'(f^{-1}(z))]^3$ for $z = y$, $f = P$:

$$\begin{aligned} D_P^2[g] &= D_y^2[g]\,(D_P[y])^2 + D_y[g]\,D_P^2[y] \\ &= \frac{D_y^2[g]}{(D_y[P])^2} \;-\; \frac{D_y[g]\,D_y^2[P]}{(D_y[P])^3}. \end{aligned} \tag{16}$$

Locally at $y = L$, $D_y[g]\big|_{y=L} = -a\,D_y[P]$ and $D_y^2[g]\big|_{y=L} = -2a\,D_y^2[P]$, so

$$\boxed{D_P^2[g]\big|_{y=L} = -\frac{2a\,D_y^2[P]}{(D_y[P])^2} + \frac{a\,D_y^2[P]}{(D_y[P])^2} = -\frac{a\,D_y^2[P]}{(D_y[P])^2}.} \tag{17}$$

Since $D_y^2[P] > 0$, the gamma of a long T-Lock ($a = +1$) is *negative* near the locked yield.

Proposition 3 (Gamma-sign threshold)**.** *For $a = +1$,*

$$\Gamma < 0 \iff y - L \;<\; \frac{D_y[P]\,D_y^3[P]}{(D_y^2[P])^2 - D_y[P]\,D_y^3[P]} \;>\; 0. \tag{18}$$

Sketch – positivity of the denominator. With $q_i = t - t_i \le 0$, $Q_i = \alpha_i e^{-(t_i - t)y} > 0$, and writing $D_y^k[A] = \sum_{t_i > t} q_i^k Q_i$, a direct expansion (Pucci Appendix C) gives

$$\big(D_y^2[A]\big)^2 - D_y[A]\, D_y^3[A] = -\sum_{i<j} q_i q_j Q_i Q_j (q_i - q_j)^2 > 0, \tag{19}$$

since $q_i q_j > 0$ (both negative) and $(q_i - q_j)^2 \ge 0$ – a Cauchy–Schwarz-type inequality on the discrete coupon measure. Combining the redemption term with the annuity inequality yields $(D_y^2[P])^2 - D_y[P]\, D_y^3[P] > 0$. □

Remark 1 (Empirical robustness)**.** *For 10Y/5Y benchmarks, the threshold* (18) *computed on 2008–2019 data sits at* $\sim 11.4\%$ *and* $\sim 21.3\%$ *respectively (paper Tables 4–5); historically observed yield moves of magnitude* $\ll$ *these bounds keep the gamma firmly negative throughout typical conditions.*

4.4 Implication: the proxy earns gamma

The trader books a short T-Lock as a long forward at strike $K = P_{t_e}(L)$ and delta-hedges against the bond. Because the T-Lock is locally *negatively*-convex in the bond's price, the linear proxy (which has zero convexity) over-hedges the position, and dynamic re-hedging realises *positive* gamma P&L (Carr–Madan 2002). Equivalently:

$$\text{P\&L}_{[t,t+dt]} \;\approx\; -\tfrac{1}{2}\,\Gamma\,(\Delta P)^2, \qquad \Gamma < 0 \;\Rightarrow\; \text{P\&L} \ge 0. \tag{20}$$

This is the formal justification for the desk practice of booking T-Locks as forwards.

5 Forward Price under Repo

5.1 Zero-haircut formula

A reverse-repo strategy delivers the security at t_e in exchange for borrowed cash. The forward price reads (simply-compounded forward repo rate r_{tu}^{repo}):

$$\text{ForwardPrice}_t(t_e) = P_t^{\text{mkt}}\big(1 + r_{t,t_e}^{\text{repo}}(t_e - t)\big) - c \sum_{t<t_i\le t_e} \alpha_i\big(1 + r_{t_i,t_e}^{\text{repo}}(t_e - t_i)\big). \tag{21}$$

5.2 Derivation via repo cash account

Following Pucci's Appendix D. The trader buys the bond at t for X_t funded by the repo (rate r^{repo}, continuously compounded), delivers it to the repo counterparty, then receives coupons c_i at $t < T_i \le T$ and uses them to pay down the borrowed cash. The repo cash account A_u evolves as

$$A_u = \sum_{t<T_i\le u} c_i\, e^{r^{\text{repo}}(u-T_i)} - X_t\, e^{r^{\text{repo}}(u-t)}, \quad u \le T. \tag{22}$$

At maturity T, the trader regains the bond worth X_T, settles $|A_T|$ with the counterparty; the net proceeds, by no-arbitrage, must equal the forward delivery, giving

$$\text{ForwardPrice}_t(T) = X_t\, e^{r^{\text{repo}}(T-t)} - \sum_{t<T_i\le T} c_i\, e^{r^{\text{repo}}(T-T_i)}. \tag{23}$$

5.3 Haircut and unsecured-funding blend

If the repo trade carries haircut $h^{\text{cut}} \in [0,1]$ and the non-collateralised portion is funded at unsecured rate r^{fun}, the blended formula is

$$\begin{aligned}\text{ForwardPrice}_t(T) = X_t\Big[(1-h^{\text{cut}})e^{r^{\text{repo}}(T-t)} + h^{\text{cut}}e^{r^{\text{fun}}(T-t)}\Big] \\ - \sum_{t<T_i\le T} c_i\, e^{(h^{\text{cut}}r^{\text{fun}}+(1-h^{\text{cut}})r^{\text{repo}})(T-T_i)}.\end{aligned} \tag{24}$$

Limits: $h^{\text{cut}} = 0$ recovers (23); $h^{\text{cut}} = 1$ replaces r^{repo} with r^{fun} throughout.

5.4 Repo sensitivity

By chain rule, with $\partial\text{ForwardPrice}_t/\partial r^{\text{repo}} > 0$ (flat short-term repo curve assumption),

$$\frac{\partial g}{\partial r^{\text{repo}}} = \underbrace{\frac{\partial g}{\partial y}\Big|_{y=R_t}}_{>0\ (\text{long})} \cdot \underbrace{\frac{\partial y}{\partial P}\Big|_{P=\text{ForwardPrice}_t}}_{<0} \cdot \underbrace{\frac{\partial \text{ForwardPrice}_t}{\partial r^{\text{repo}}}}_{>0} \;<\; 0. \tag{25}$$

Higher repo $\Rightarrow$ richer forward price $\Rightarrow$ lower forward IRR $\Rightarrow$ cheaper long T-Lock. *Specialness* on the on-the-run benchmark (where repo rates can drop several hundred basis points below GC, as during the 2014 squeeze) makes the long T-Lock significantly more expensive.

5.5 EUR vs USD repo regimes

In a negative-repo regime (EUR), (23) produces a forward price below spot (after coupon adjustment), inflating long-T-Lock value relative to the analogous USD trade.

6 The Then-Current T-Lock and Roll P&L

6.1 Booking convention

For the Then-Current variant, the underlying security is the on-the-run benchmark at t_e – unknown at trade date. The desk practice:

1. At inception, book the trade as a forward on the current on-the-run benchmark.
2. At each new-issue auction date t^{roll}, revise the booking to use the new on-the-run security; the booking change generates a discrete *roll P&L*.

6.2 Roll P&L derivation

At the switch from coupon c to coupon $\hat{c}$ (all other schedule details unchanged), and writing $\hat{R}_t$ for the new security's t_e-forward IRR vs. R_t for the old one:

$$K = P_{t_e}(L), \qquad \hat{K} = \hat{P}_{t_e}(L), \tag{26}$$

$$\pi_t = D_{t,t_e}\big[K - F_t(t_e)\big], \qquad \hat{\pi}_t = D_{t,t_e}\big[\hat{K} - \hat{F}_t(t_e)\big], \tag{27}$$

where $F_t(t_e) = P_{t_e}(R_t)$ and similarly for $\hat{F}$. The forward P&L of the switch is

$$\frac{\hat{\pi}_t - \pi_t}{D_{t,t_e}} \;=\; (\hat{K} - K) - \big(\hat{F}_t(t_e) - F_t(t_e)\big). \tag{28}$$

Computing each piece (and using the closed-form (4)):

$$\hat{K} - K = (\hat{c} - c) \sum_{t_i > t_e} \alpha_i e^{-(t_i - t_e)L}, \tag{29}$$

$$\hat{F}_t(t_e) - F_t(t_e) = (\hat{c} - c) \sum_{t_i > t_e} \alpha_i e^{-(t_i - t_e)\hat{R}_t} + \big[P_{t_e}(\hat{R}_t) - P_{t_e}(R_t)\big]. \tag{30}$$

Substituting:

$$\boxed{\frac{\hat{\pi}_t - \pi_t}{D_{t,t_e}} = (\hat{c} - c)\big[A_{t_e}(L) - A_{t_e}(\hat{R}_t)\big] + \big[P_{t_e}(R_t) - P_{t_e}(\hat{R}_t)\big],} \tag{31}$$

with $A_t(y) = \sum_{t_i > t} \alpha_i e^{-(t_i - t)y}$.

6.3 First-order approximation

Linearising A_{t_e} at $\hat{R}_t$ and P_{t_e} at R_t:

$$\begin{aligned} \frac{\hat{\pi}_t - \pi_t}{D_{t,t_e}} \approx\ & (\hat{c} - c)(L - \hat{R}_t)\, D_y[A_{t_e}](\hat{R}_t) \\ & - (\hat{R}_t - R_t)\, D_y[P_{t_e}](\hat{R}_t). \end{aligned} \tag{32}$$

For short tenors $t_n - t_e$ small, this reduces to (paper eq. 70–73):

$$\frac{\hat{\pi}_t - \pi_t}{D_{t,t_e}} \approx (\hat{c} - c)(\hat{R}_t - L) \sum_{t_i > t_e} \alpha_i (t_i - t_e) + (\hat{R}_t - R_t) \left[(t_n - t_e) + c \sum_{t_i > t_e} \alpha_i (t_i - t_e) \right]. \tag{33}$$

6.4 Conditions for negligible roll P&L

From (31)–(33), the roll P&L vanishes if *any* of:

1. $\hat{c} = c$ (no coupon change at re-issue);
2. $\hat{R}_t = R_t$ (new on-the-run prices to the same forward IRR);
3. $L = R_t$ *and* $\hat{c} = c$ (ATM trade with same coupon).

The auction process (Pucci Fig. 4) is precisely structured so that the new benchmark prices at par at the prevailing IRR; deviations come from auction-day demand/supply effects on $\hat{c}$ and the $\hat{R}_t - R_t$ gap. Empirically these are small for liquid benchmarks (2Y, 5Y, 10Y).

7 Statistical Findings (Paper §4)

The paper validates the proxy on 2008–2019 daily data for 2Y, 5Y, 10Y US Treasury benchmarks.

Overhedge error, 6M T-Lock on 10Y	avg.	st.dev.	5%-ile	95%-ile
$y - L$ (yield change)	−0.090%	0.663%	−1.44%	+0.93%
$(y - L)^2$	0.004%	0.006%	~ 0	0.0206%
Negative-gamma upper bound (eq. 18)	avg.	st.dev.	5%-ile	95%-ile
10Y benchmark	11.44%	0.10%	11.28%	11.61%
5Y benchmark	21.31%	0.12%	21.11%	21.51%
2Y benchmark	22.47%	15.20%	11.15%	25.57%
Roll P&L (10Y, $L = 3\%$, trader view)	1M	3M	6M	12M
average	−0.108%	−0.107%	−0.105%	−0.100%
95%-ile (worst)	−1.04%	−1.02%	−1.00%	−0.95%

Headlines: (i) the overhedge error is $\sim 0.32\%$ of notional on average on the 6M 10Y, $\sim 1.55\%$ at the 95%-ile (the latter realised during the August 2011 US downgrade); (ii) the gamma never flips sign in the sample for the 5Y/10Y; (iii) the average roll P&L is small but *adverse* to the trader – the 10Y benchmark roll consistently moves the IRR upward at issuance, generating a $\sim$ 10bp loss per roll on notional.

8 Counterparty and Funding Adjustments (Paper §5)

A T-Lock book carries the standard derivatives valuation adjustments:

- **CVA / DVA.** Short-tenor + corporate exposure – generally small but with *systemic wrong-way risk*: a flight-to-quality regime simultaneously rallies Treasuries (raising the bank's exposure to the client) and weakens corporate credit. Pucci argues the bank's own DVA typically moves in offset.
- **FVA.** Funding cost arises from the cash leg of the repo hedge, plus collateral-funding under CSA/cleared repo (Andersen, Duffie & Song 2019).
- **KVA.** Capital cost is minimal due to the short tenor.
- **Proxy underestimation.** Booking via the linear forward overhedges the trade and *underestimates* the position's exposure – hence the contra-XVA contributions are also underestimated, but mildly so for short-dated ATM trades.

9 Validation Notes for the Pricing Library

9.1 Bond-price machinery

1. Implement (2) (full T-Lock prescription) and (3) (stub period). Cross-check against the closed-form annual 30/360 case $P = (1+R)^{-n} + \frac{c}{R}(1-(1+R)^{-n})$.
2. Implement Hull-form (4) for greek analysis; verify the derivative formula (5) via finite differences at multiple yield levels.
3. Robust IRR root-finding: Newton from $y_0 = c$ for non-distressed issues; bisect-fallback when $|D_y[P]|$ is very small.

9.2 Forward price

4. Implement (21) (simply-compounded, zero haircut). Smoke-test: with no coupons in $[t, T]$, $\text{ForwardPrice} = P^{\text{mkt}}(1 + r^{\text{repo}}\tau)$.
5. Implement (24) for haircut/funding split; verify the limits $h^{\text{cut}} = 0$ and $h^{\text{cut}} = 1$.
6. Verify (25): bumping r^{repo} +10bp should reduce a long-T-Lock present value.

9.3 T-Lock pricing

7. Booking value $v_t = a\, D_{t,t_e}\,(K - \text{ForwardPrice}_t(t_e))$ with $K = P_{t_e}(L)$. ATM smoke-test: $\text{IRR}_t = L$ at trade date $\Rightarrow v_t \approx 0$, residual is exactly the overhedge premium.
8. Independent MC under a short-rate / yield model: simulate IRR_{t_e} under $\mathbb{Q}^{t_e}$, evaluate (6); the proxy should overhedge by $\sim \frac{1}{2} M \sigma_y^2 t_e$.
9. Reproduce paper's Figure 1: T-Lock payoff alongside its forward overhedge as functions of P_{t_e}.

9.4 Greeks and replication

10. Risk factor / DV01: RiskFactor $= -10{,}000 \cdot \mathrm{DV01}$, $\mathrm{DV01} = (P(R + 1\mathrm{bp}) - P(R - 1\mathrm{bp}))/2$. Verify against the analytic $D_y[P]$ from (5).
11. Delta sign: confirm $D_P[g] < 0$ for long T-Lock at $|y - L|$ small; confirm sign-flip occurs only above the threshold of Lemma 1.
12. Gamma sign: confirm $D_P^2[g] < 0$ at $y = L$; confirm the local expression (17); reproduce 10Y bound $\sim 11.4\%$ from (18).
13. Replication P&L: simulate daily Treasury yields, dynamically re-hedge $\Delta = D_{t,t_e} D_P[g]$ via repo, accumulate P&L. Verify the realised P&L is non-negative and approximately $-\frac{1}{2}\int \Gamma_t \, (d\mathrm{IRR}_t)^2$.

9.5 Then-Current and roll P&L

14. Implement (31) as the full closed-form roll P&L. Smoke-test: with $\hat{c} = c$ and $\hat{R}_t = R_t$, P&L vanishes.
15. Implement (33) as a fast first-order check. Reproduce paper Tables 7–10 by running the formula on a historical auction calendar.
16. At-par new issue check: solve $\hat{c}$ from $P_t^{(\hat{c})}(R_t) = 1$, then $\hat{R}_t = R_t$, and verify roll P&L $= 0$.

10 Mathematical Appendix Recap (Pucci §A–C)

For completeness, the technical lemmas used in the gamma proof:

$$(f \circ g)' = (f' \circ g) \cdot g', \qquad (f \circ g)'' = (f'' \circ g) \cdot (g')^2 + (f' \circ g) \cdot g'', \tag{34}$$

$$(f^{-1})' = 1/f', \qquad (f^{-1})'' = -f''/(f')^3, \tag{35}$$

and the Cauchy–Schwarz-type identity (19), central to the gamma-sign analysis.

Reference. Pucci, M. (2019). *Hedging the Treasury Lock.* Working paper, Banca IMI SpA, Largo Mattioli 3, Milan, Italy. SSRN no. 3386521.

Companion papers. M. Pucci (2012a), *Constant Maturity Asset Swap Convexity Correction*, SSRN 1961545; M. Pucci (2012b), *Pricing Index-Linked Hybrids under a Local Volatility Market Model*, SSRN 2056277; M. Pucci (2014), *Constant Maturity Treasury Convexity Correction*, IJTAF 17(8), SSRN 3387961.